QP CODE: **Reg No:.....**

Name:.....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR EXAMINATION OCTOBER 2026

Fifth Semester

MG5DSECFS305 - Biometric Security

(2024 ADMISSION ONWARDS)

Duration: 2 Hours Maximum Marks: 70

Remember (K), Understand (U), Apply (A), Analyse (An), Evaluate (E), Create (C), Interest (I), Appreciation (Ap), and Skill (S)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Very Short Answer Questions

Answer any **10** questions.

Each question carries **2** marks

1. Define biometrics. [K] [CO1]

2. Distinguish between biometric techniques and traditional techniques. [U] [CO1]

3. Differentiate between verification and identification. [U] [CO1]

4. Explain False Acceptance Rate (FAR) and its significance in biometric security. [U] [CO1]

5. Analyse any two characteristics that make fingerprints suitable for biometric identification. [An] [CO2]

6. Describe the basic purpose of facial recognition. [U] [CO2]

7. Illustrate how iris recognition can be used for biometric identification. [A] [CO2]

8. Explain the concept of retina vascular pattern recognition in biometrics. [U] [CO3]

9. Examine any two characteristics used in hand scanning for biometric identification. [An] [CO3]

10. Discuss DNA biometrics. [U] [CO3]

11. Illustrate how keystroke dynamics can be used for user authentication. [A] [CO4]

12. Mention any two characteristics of voice biometrics. [U] [CO4]

(2 × 10 = 20)

Part B

Short Answer Questions

Answer any **6** questions.

Each question carries **5** marks

13. Discuss Biometric Matching. [U] [CO1]

14.Differentiate False match rate and false nonmatch rate. [U] [CO1]

15. Discuss the components of finger scan technology. [U][CO2]

16. Analyse the different stages involved in facial scan technology. [An] [CO2]

17. Analyse the strength and weakness of Retina vascular Technology. [An][CO3].

18. Illustrate how the characteristics of Hand Scan Technology can be applied in a biometric authentication system. [A][CO3]

19. Evaluate the effectiveness of keystroke dynamics as a biometric authentication technique. [E][CO4]

20. Explain the deployment of Voice Scan Technology in biometric systems. [U][CO4]

(5 × 6 = 30)

Part C

Essay Questions

Answer any **2** questions.

Each question carries **10** marks

21.Discuss competing finger scan technology. [U][CO1]

22.Discuss Strength and weakness of iris scan Technology. [U][CO3]

23. Analyse the characteristics, strengths, and weaknesses of Voice Scan Technology. [An][CO4]

(10 × 2 = 20)